

[Kat]herine Titterton

San Francisco, CA · macrofluidic.me · github.com/kat-titter · X @KatTitterton

TL;DR Scientist-engineer at the seam of imaging, machine learning, and lab automation in biotech. 8+ years turning bench work into machine-learning fuel: building and scaling high-content assays, then the image analysis that unlocks phenotypic discovery at scale.



[MACROFLUIDIC.ME/RESUME](https://macrofluidic.me/resume)

100,000+

compounds across two of the world's largest Cell Painting datasets

~10×

more sensitive bioactivity detection than conventional toxicity assays

0.92 AUC

in silico drug-induced liver injury risk model, co-developed

8 papers

peer-reviewed, incl. Nat Commun x2, Cell Syst, J Biol Chem

EXPERIENCE

publications listed on page 2

Founding Member of the Technical Staff · Axiom Bio

2023–present · San Francisco, CA

- Oversaw assay development and data generation for two of the world's largest Cell Painting datasets (100,000+ compounds), the training data for in silico toxicology models.
- Invented two bioactivity-detection methods ~10× more sensitive than conventional toxicity assays (LDH, MT-Glo, cell count).
- Led feature engineering and algorithm development as founding technical hire; co-developed a 0.92-AUC DILI clinical-risk model (SOT 2025). Honorary Artist in Residence.

Nat Commun 2026 · Cell Syst 2026

Associate Scientist III · Insitro, Arrayed Screening, Neuroscience

2022–2023 · South San Francisco, CA

- Scaled high-content phenotypic assays to high-throughput small-molecule screens (TSC, ALS) on near-fully-automated platforms (BioTek, High-Res).
- Built a label-free digital phase-contrast timecourse modality and Python analysis suite to track neuron growth dynamics.

Technical Associate II · MIT, Synthetic Neurobiology Group

2021–2022 · Cambridge, MA

- Imaged 30+ proteins at nanoscale by optimizing multiplexed expansion revealing (mExR) for 7+ IHC rounds on 16× expanded tissue.
- Built image-analysis pipelines in ImageJ/Fiji and MATLAB.

Nat Commun 2024

Associate Scientist II · AbbVie, Foundational Neuroscience Center

2018–2021 · Cambridge, MA

- Built primary murine and hiPSC neuronal tau-aggregation high-content screening assays (Revvity Opera, Harmony) used to rank lead candidates.
- Modeled tau propagation in Alzheimer's disease with microfluidic devices; authored R pipelines for reproducible dose-response profiling.

Mol Neurodegener Adv 2026 · iScience 2026 · J Biol Chem 2025 · iScience 2022

Research Technician I · Massachusetts General Hospital

2017–2018 · Boston, MA

- Elucidated mechanisms of general anesthetics with voltage-clamp electrophysiology in GABAR-expressing *Xenopus* oocytes.

Eur J Med Chem 2019

PROJECTS

Bare Metal · First IRL art exhibit, Human Tech Week, SF; post-hoc analysis on macrofluidic.me/blog

2025

Vivarium 2.0 · Rebuilt poison-dart-frog habitat for Axiom Bio: automated misting, motion-triggered live video

2024

OpenTransformers · LLM code-generation companion for the OT-2 lab robot; 2nd place, BioHack NYC

2023

EDUCATION

Harvard University · Extension School, graduate coursework

2017–2020

Graduate coursework while employed full-time: Python programming; image processing & computer vision; social medicine.

B.A., Colgate University · *cum laude*

2013–2017

Cellular Neuroscience major, Mathematical Systems Biology minor. Co-captain, Division I Track & Field; Dorothy-Waltham Award for Athletic Leadership.

SKILLS & TOOLS

ML & DATA

Python (NumPy, SciPy, scikit-learn, pandas) · image embeddings: clustering, feature derivation, normalization · multivariate statistics · design of experiments · LLM codegen (Claude Code, Cursor)

AUTOMATION

Liquid handlers & schedulers: OpenTrons, Integra, BioTek, Labcyte Echo, High-Res Biosolutions, Green Button Go · Benchling ELN

IMAGING & ANALYSIS

High-content & confocal imaging (Opera Phenix, Operetta) · Harmony · ImageJ/Fiji macros · image segmentation · dose-response fitting (Prism; R: drc, platetools)

WET LAB

Primary murine & hiPSC neuron culture · microfluidic models · ICC/IHC · assay development & HT/HC screening · stereotaxic surgery, microdissection · voltage-clamp electrophysiology

CERTIFICATIONS

FII Level 1 Freediving (Oct 2019) · Molchanov L2 Freediving (Oct 2025) · SII Shark Ecology Specialty (Apr 2020)

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PEER-REVIEWED PUBLICATIONS

8 papers · newest first

1. Yanamandra K, Romanul N, Patel H, Sanders H, Chang R, Darcy A, Brown N, Kane C, Wu HY, Jagtap K, Chancellor S, Morozova O, Xi M, Hogan C, **Titterton K**, Wu JW, Manos J, Karran E, Dellovade T, Langlois X. Conformation-specific anti-pathological tau antibody raised against AD brain-derived seed-competent tau reduces tau pathology in transgenic mice. *Mol Neurodegener Adv*. 2026;2(1):43. doi:10.1186/s44477-026-00049-6
2. Sebastian RM, Kupp R, Nanavati D, Parfentev I, Krzystek TJ, Romanul N, Yanamandra K, Preiss CN, Manos JD, Wu HY, Volkova A, Crescenti S, **Titterton K**, Welker AM, Townsend M, Karran E, Langlois X, Wu JW. SynCAR platform for capturing neuronal synaptic proteopathic seeds. *iScience*. 2026;29(7):116362. doi:10.1016/j.isci.2026.116362 PMID 42325569 · PMC13276770
3. Ewald JD, **Titterton KL**, Bäuerle A, Beatson A, Boiko DA, Cabrera AA, Cheah J, Cimini BA, Gorissen BL, Harrill J, Jones TR, Karczewski KJ, Crute CE, Rouquie D, Seal S, Weisbart E, White B, Carpenter AE, Singh S. Cell Painting for cytotoxicity and mode-of-action analysis in primary human hepatocytes. *Cell Syst*. 2026;17(5):101566. doi:10.1016/j.cels.2026.101566 PMID 41903531
4. Seal S, Dee W, Shah A, Cerisier N, Zhang A, Miglietta E, **Titterton K**, Cabrera AA, Boiko D, Beatson A, Slabaugh G, Taboureaux O, Carreras Puigvert J, Singh S, Spjuth O, Bender A, Carpenter AE. Counting cells can accurately predict small-molecule bioactivity benchmarks. *Nat Commun*. 2026;17(1). doi:10.1038/s41467-026-68725-5 PMID 41651839 · PMC12988037
5. Wu JW, **Titterton K**, Sebastian R, Romanul N, Yanamandra K, Mastis B, Chang R, Khasnavis S, Kwon T, Chai Y, Manos JD, Preiss CN, Hu M, Welker AM, Liao F, Dellovade T, Karran E, Langlois X. Development of a robust, physiologically relevant neuronal tau aggregation assay for tau-related drug discovery. *J Biol Chem*. 2025;301(12):110853. doi:10.1016/j.jbc.2025.110853 PMID 41135680 · PMC12670552
6. Kang J, Schroeder ME, Lee Y, Kapoor C, Yu E, Tarr TB, **Titterton K**, Zeng M, Park D, Niederst E, Wei D, Feng G, Boyden ES. Multiplexed expansion revealing for imaging multiprotein nanostructures in healthy and diseased brain. *Nat Commun*. 2024;15(1):9722. doi:10.1038/s41467-024-53729-w PMID 39521775 · PMC11550395
7. Manos JD, Preiss CN, Venkat N, Tamm J, Reinhardt P, Kwon T, Wu J, Winter AD, Jahn TR, Yanamandra K, **Titterton K**, Karran E, Langlois X. Uncovering specificity of endogenous TAU aggregation in a human iPSC-neuron TAU seeding model. *iScience*. 2022;25(1):103658. doi:10.1016/j.isci.2021.103658 PMID 35072001 · PMC8761709
8. Wu B, Jayakar SS, Zhou X, **Titterton K**, Chiara DC, Szabo AL, Savechenkov PY, Kent DE, Cohen JB, Forman SA, Miller KW, Bruzik KS. Inhibitable photolabeling by neurosteroid diazine analog in the $\beta 3$ -subunit of human heteropentameric type A GABA receptors. *Eur J Med Chem*. 2019;162:810-824. doi:10.1016/j.ejmech.2018.11.019 PMID 30544077 · PMC6912599

CONFERENCE PRESENTATIONS & POSTERS

1. **Titterton K**, Boiko D, Cabrera AA, Bäuerle A, Tyagi A, Saneinejad S, Beatson A, White B. "Accurate and interpretable in silico clinical risk assessment for drug-induced liver injury (DILI) from molecular structure." *Society of Toxicology (SOT) Annual Meeting*, 2025.
2. **Titterton K**, et al. "Development of a tau propagation neuronal microfluidic cell culture model for assessing inhibitors of tau propagation." *Society for Neuroscience (SfN) Global Connectome Virtual Meeting*, January 2021.
3. **Titterton K**, et al. "High content analysis (HCA) of tau aggregation in primary neurons: screening anti-tau antibodies and modeling tauopathy progression in vitro." *SBI2 High Content Meeting*, September 2019, Harvard Medical School, Boston, MA.

Earlier research presentations

- "Investigating RNAi as a potential therapeutic for Huntington's Disease in a transgenic mHTT145Q zebra finch (*Taeniopygia guttata*) model." Colgate University Neuroscience Thesis Poster Presentations, 2017.
- "Atopy and helminth infection in Ethiopian schoolchildren." Colgate Summer Research Poster Symposium, 2016.
- "PDE2 regulation of β -arrestin signaling." Systems Biology Center New York (SBCNY) Summer Trainee Poster Symposium, Icahn School of Medicine at Mount Sinai, 2015.